QUESTIONS 5 / 5 completed

Quick Sort

QUICK SORT

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[], int low, int high)
{
    int pivot = arr[high];
    int i = low - 1;

    for (int j = low; j < high; j++)
    {
        if (arr[j] < pivot)
        {
            i++;
            swap(&arr[i], &arr[j]);
        }
```

INPUT

6
50 23 9 18 61 32

QUESTIONS 5 / 5 completed

Bubble Sort

BUBBLE SORT

"Write a program that implements the bubble sort
"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    int n, i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    // Bubble Sort
    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
```

INPUT

6
50 23 9 18 61 32

QUESTIONS 5 / 5 completed

Insertion Sort

INSERTION SORT

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    int arr[] = {9, 5, 1, 4, 3};
    int n = 5, i, key, j;

    for (i = 1; i < n; i++)
    {
        key = arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }

```

INPUT

9 5 1 4 3

QUESTIONS 5 / 5 completed

Merge Sort

MERGE SORT

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

void merge(int arr[], int left, int mid, int right)
{
    int n1 = mid - left + 1;
    int n2 = right - mid;

    int L[n1], R[n2];

    for (int i = 0; i < n1; i++)
        L[i] = arr[left + i];

    for (int j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];

    int i = 0, j = 0, k = left;

    while (i < n1 && j < n2)
    {
        if (L[i] <= R[j])
            arr[k++] = L[i++];
```

INPUT

7
38 27 43 3 9 82 10